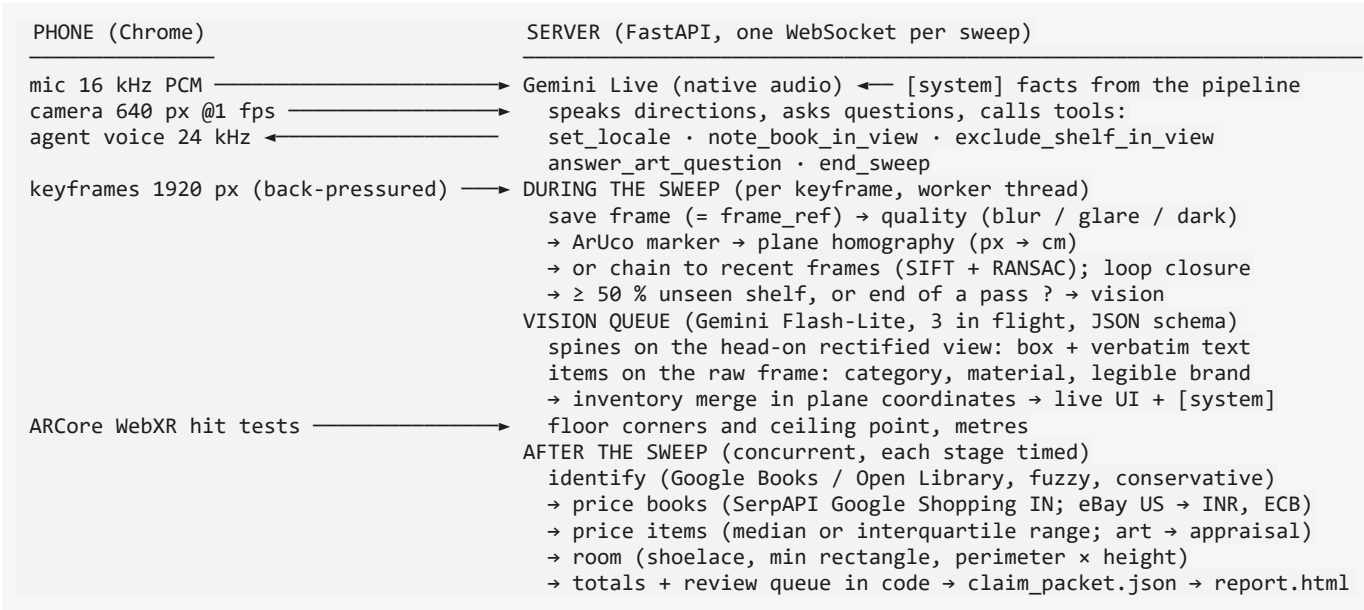


Architecture



Which model does what

Job	Model / tool	Why
Conversation, directing the sweep	Gemini Live native-audio model	real-time voice in and out, barge-in, tool calls
Spine boxes and transcription	Gemini Flash-Lite on the rectified shelf view	reads small rotated text; boxes come back on a 0–1000 grid; strict "copy, don't guess" schema
Non-book items	Gemini Flash-Lite on raw frames	category, material, legible brand; never prices
Everything numeric	plain code	homographies, medians, areas and totals are deterministic and tested

Where metric scale comes from

- **Spines.** An ArUco marker of known, tape-measured size, shown on a laptop or printed, sits on each shelving unit. Its corners give a homography from image pixels to centimetres on the shelf plane.
- Frames are re-projected head-on before spines are detected, so a box's width divided by the resolution (px per cm) is the true thickness.
- Measuring boxes in the raw, oblique photo overstated a 3.2 cm spine as 7.3 cm. Rectification brought it to within 5%, and `tests/test_scale.py` checks this.
- Frames without the marker inherit scale by chaining homographies to a frame that has it. That's exact for a plane, and the chain stops at a weak link rather than drifting.
- **Room.** ARCore motion tracking via WebXR hit tests gives metric 3D points in one world frame during the same walk.

Where prices come from

- **Every figure is traceable.** It's the median of matched listings returned by SerpAPI (Google Shopping in the claim country, eBay for used copies). Each one carries its merchant, URL, retrieval time and the rule used to pick it.
- **Conversions are labelled.** Converted figures carry the ECB rate and its date (frankfurter.dev).
- **The model never produces a price.** With no listing, the amount stays empty, the line goes to the review queue and it's excluded from totals.
- **Raw responses are kept.** Every API response is saved in `sweeps/<id>/raw/` for audit.